

# Project U.W.B.O.T.H.A.N.

(Ultra-Wideband Operational Transmission Hub for Allied Networks)

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## Group 11:

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**Department of Computer Science and Engineering**

**Wright State University**

**Date: 4/9/2026**

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# 1. Overview

Our rebel alliance needs a covert way of transmitting crucial images of the Death Star to a rebel server. They have managed to board the Death Star and are unable to use traditional means of data transmission such as Wi-Fi or Bluetooth. The data must be encrypted when sent, and it must confirm at both ends of the transmission process that all data is there. Once the data has been sent and received it will be parsed through and uploaded to a rebel web application, where they can view the Death Star images, and with additional two factor authentication, the rebels can view a video. A Raspberry Pi 4B will be provided, alongside a USB stick with 100 images and 10 of those 100 will be our target images to detect and transmit.

## 1.1 Rationale

The approach our group has chosen is Ultra-wideband (UWB), which is wireless technology that uses radio waves for location tracking and data transmission. This choice covers both the covert side of things, as this technology can be completely undetectable, and its speed and throughput will be reliable. An alternative solution that was considered was via QR codes, which was ultimately passed on due to its slow speed and complexity. Rather, UWB offered a practical solution that uses neither Wi-Fi nor Bluetooth, can send encrypted data, and allows for bidirectional communication between the transmitter and receiver to ensure our data has not been altered.

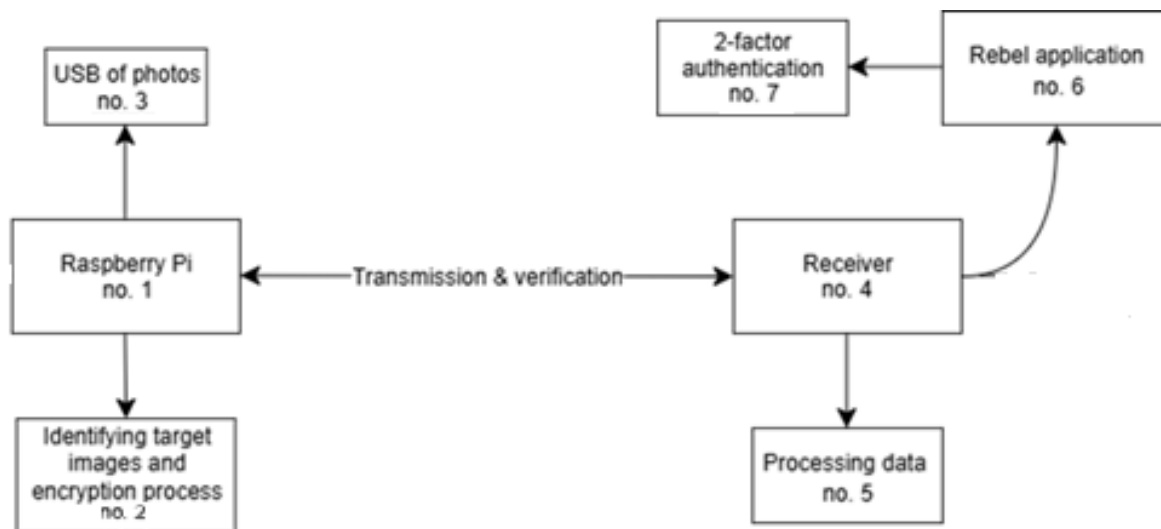
## Project Pivot

Due to hardware availability issues, we are unable to implement UWB for the project. Moving forward we will be implementing a modified design that uses Ethernet instead of UWB radio.

Ethernet provides fast, reliable bidirectional communication that meets all of the requirements of this project outside of the need for a wireless solution.

## 1.2 Design

Starting with the hardware, the Raspberry Pi 4B will be our main device that will host the USB stick of images. The Raspberry Pi 4B will be connected to a laptop via Ethernet cable. This will allow the UWB module to receive data and transfer it to the laptop. On the software side of things, the Raspberry Pi 4B will contain all the code to filter the images, crop the images, encrypt the data, calculate a hash using sha512, and transfer it to the laptop via Ethernet. On the receiving side of things, there will be a script on the laptop to accept and wait for the data to be transmitted from the Raspberry Pi 4B, this script will also use sha512, to ensure all data was sent. If data is missing it will request another send, otherwise it will send the data/images to a Fast API backend that will be running on the laptop to upload them to an angular based web application. The backend will also be where the two-factor authentication functionality will live.



**Figure 1:** Diagram depicting the overall functionality of the design

**Source:** Author

## **1.3 Analysis of Major Subsystems**

Project U.W.B.O.T.H.A.N. is made up of 5 main subsystems: the image targeting and hashing software on the Raspberry Pi, the Ethernet connection of the Pi to laptop, the communication between transmitter and receiver, processing/validation on the receiving side, and the backend/frontend web application. The Raspberry Pi is responsible for the image targeting and hashing software, which is compatible, as the Pi is designed to allow programming. As for processing the data, that will be done through a script running on the laptop which will be sent from the receiver UWB module through the USB cable to the laptop. During that process, they backend and frontend will be running, waiting for data to be captured to display them to the user.

## **1.4 Safety Risks**

Our project offers low minimal safety risks, as all hardware involved is low-power components. The main risks to be aware of include cuts from any protruding sharp objects from the hardware, short circuiting components, and burns from overheating. To avoid these risks, the following will be done; inspecting the hardware and carefully handling it, ensuring all connections are properly wired, and allowing all devices to have access to open spaces to enable proper air flow.

## **1.5 Security Risks**

Failure to secure any data could result in imperial guards intercepting the data and ruining the mission. Security is a top priority to ensure all data is encrypted, and only the rebel hardware will be able to decrypt the data and view the contents. Even though the guards still could intercept the message, they will be unable to decrypt the data due to the complicated nature of the encryption

process that is AES. Another precaution that is being taken is data integrity. By utilizing sha512 our data will be checked that none was lost or tampered with during the transmission process, failure to implement could result in the imperial guards intercepting and providing false data for us rebels. Finally, the use of Ethernet is not covert like UWB is. However, good security can still be maintained through AES encryption and sha512 hashing.

## 2. Requirements, Standards, and Constraints

### 2.1 Requirements

| Req No. | Obj No. | Requirement                                                                                                                                                        |
|---------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10      | 1       | The transmitter and receiver devices shall be under [Equation] in volume.                                                                                          |
| 20      | 1       | The transmission process shall meet at least 6 Mbps.                                                                                                               |
| 30      | 2       | *Two ultra-wideband devices will be used, one being the transmitter and the other being the receiver, and the total for both shall fall within the price objective |
| 40      | 4       | The transmission process shall be encrypted using AES.                                                                                                             |
| 50      | 5       | *The frequency ranges Project U.W.B.O.T.H.A.N will utilize will be of 3.1 to 10.6 GHz                                                                              |
| 60      | 3       | The Raspberry Pi will utilize OpenCV to detect the red circles in the target images.                                                                               |
| 70      | 6       | Project U.W.B.O.T.H.A.N. will use angular to support its application that will host the target images.                                                             |
| 80      | 5       | *Project U.W.B.O.T.H.A.N. will utilize ultra-wideband (UWB) to achieve wireless data transmission, from receiver to transmitter and vice-versa.                    |

**Figure 2:** Table of Requirements      **Source:** Author

**\*These requirements have been superseded due to the transition from UWB to Ethernet.**

## 2.2 Standards

| Stand. No | Standard                                                              |
|-----------|-----------------------------------------------------------------------|
| 10        | Encryption - FIPS 197, Advanced Encryption Standard (AES)             |
| 20        | Hashing – FIPS 180-4 Secure Hash Standard                             |
| 30        | Project requirements - IEEE 1012 Software Verification and Validation |
| 40        | IEEE 12207 Software Life Cycle Processes                              |

**Figure 3:** Table of Standards      **Source:** Author

## 2.3 Constraints

| Const No. | Constraint                                                   |
|-----------|--------------------------------------------------------------|
| 10        | The Raspberry Pi will not use more than 25 watts of power.   |
| 20        | The Raspberry Pi will not produce more than 30 dB of sound.  |
| 30        | The file format of the images will be Jpeg only.             |
| 40        | The device will not exceed 12.5 cm <sup>3</sup> in volume.   |
| 50        | The receiver shall only be used with a windows-based device. |

**Figure 4:** Table of Constraints      **Source:** Author

## 3. Plan

### 3.1 Test Plan

#### Requirements Tests

| Requirement | Test Method            | Evaluation Method                                                                     | Threshold                             | Objective                            |
|-------------|------------------------|---------------------------------------------------------------------------------------|---------------------------------------|--------------------------------------|
| 10          | Measurement            | Physical Measurement with ruler                                                       | 12.5 cm3                              | 10.0 cm3                             |
| 20          | TransmissionSpeed test | Does the test speed meet or exceed the threshold.                                     | 6 Mbps                                | >=6.5 Mbps                           |
| 30          | Budget Log             | Maintain log of all expenses                                                          | <\$300                                | <\$300                               |
| 40          | Hands-on decryption    | A decryption test to attempt to decrypt the data using an expected and unexpected key | Successfully decrypt encrypted images | All 10 images successfully encrypted |
| 50          | Measurement            | Utilizing a device that will measure the frequency                                    | Frequency between 3.1 and 10.6 GHz    | 6.5 to 9 GHz                         |
| 60          | Software Test          | Use images with/without red circles and ensure red circle detection is working        | 98% accuracy                          | 100% accuracy                        |
| 70          | Visual                 | Web app hosted and developed in angular                                               | Web app hosted                        | Completed fully                      |
| 80          | Data Transmission      | Data sent and received.                                                               | Some form of data sent and received   | Completed fully                      |

**Figure 5:** Requirement Tests

**Source:** Author

## Standard Tests

| Standard | Test Method | Evaluation Method                                                                                                                                                 | Threshold                                             | Objective                            |
|----------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|--------------------------------------|
| 10       | Decryption  | Ensure that the encryption algorithm that is used is AES.                                                                                                         | Successfully decrypted the encrypted images with AES. | All 10 images successfully encrypted |
| 20       | Review      | Ensure that the hashing algorithm used is sha512.                                                                                                                 | sha512                                                | sha512                               |
| 30       | Review      | Ensure the transmission process has fail-safes in case the transmission process fails to re-attempt the transmission. It will utilize a lighter set of V&V tasks. | Integrity level 1 compliance                          | Integrity level 1 compliance         |
| 40       | Review      | Ensure that we are following the software life cycle process in our project development.                                                                          | 75% Compliance                                        | 100% Compliance                      |

**Figure 7: Standard Tests**      **Source:** Author

## Constraint Tests

| Constraint | Test Method | Evaluation Method                              | Threshold           | Objective       |
|------------|-------------|------------------------------------------------|---------------------|-----------------|
| 10         | Measurement | Physical measurement with a multi-meter        | Max of 25 watts     | Max of 20 watts |
| 20         | Measurement | Physical Measurement with infrared thermometer | Max of 85 degrees C | 70 degrees C    |
| 30         | Measurement | Physical Measurement with sound level meter    | Max of 30 dB        | Max of 25 dB    |
| 40         | File Type   | File Type                                      | .jpeg               | .jpeg           |
| 50         | Measurement | Physical measurement                           | 12.5 cm3            | 10.0 cm3        |

**Figure 6: Constraint Tests**      **Source:** Author

## Summary Tables of Test Results

| Requirement | Test Date | Test and Evaluation Location                                                                                              | Result               | Notes                                                                                                                                                                   | Date Passed (Accepted) |
|-------------|-----------|---------------------------------------------------------------------------------------------------------------------------|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| 10          | 4/2/2026  | Measured Raspberry Pi (our new transmitter).                                                                              | 91.4 cm <sup>3</sup> | This value is over our requirement value; however, it is still accepted due to hardware complications and having to use the Pi as our transmission device and ethernet. | 4/2/2026               |
| 20          | 4/7/2026  | 10 Tests of the thruput were taken and averaged in the lab below Russ. Code was developed to track the tranmission speed. | 44.25334 Mbps        | Our average speed exceeded our requirements. There was also variations between speeds, due to the TCP connection start up times and file size                           | 4/7/.2026              |
| 30          | 4/2/2026  | Budget Log                                                                                                                | \$0                  | Due to not getting our parts the whole project resulted in \$0.                                                                                                         | 4/2/2026               |
| 40          | 4/7/2026  | Hands-on decryption test by manually passing in our .pkg file with expected and unexpected key.                           | Pass                 | Successfully decrypts .pkg with correct key.                                                                                                                            | 4/7/2026               |
| 50          | Untested  | Did not get our UWB modules, therefore do not have a frequency to measure                                                 | Untested             | No modules                                                                                                                                                              | 4/7/2026               |
| 60          | 4/7/2026  | Red circle detection successfully detects the target images 100% of the time.                                             | Pass                 | After several tests, the detection was correct every time.                                                                                                              | 4/7/2026               |

| Requirement | Test Date | Test and Evaluation Location                                                              | Result | Notes                                                                                               | Date Passed (Accepted) |
|-------------|-----------|-------------------------------------------------------------------------------------------|--------|-----------------------------------------------------------------------------------------------------|------------------------|
| 70          | 1/5/2026  | By visual inspection upon the web app's creation, we are using angular.                   | Pass   | Web application was created via command line using Node.js/angular package                          | 4/7/2026               |
| 80          | 4/7/2026  | All tests were done in the lab and resulted in successful transmissions 100% of the time. | Pass   | All testing resulted in successful data transmission from Pi to the laptop to populate the web app. | 4/7/2026               |

**Figure 8:** Requirements Test Results **Source:** Author

| Standard | Test Date | Test and Evaluation Location | Result | Notes                                                                       | Date Passed (Accepted) |
|----------|-----------|------------------------------|--------|-----------------------------------------------------------------------------|------------------------|
| 10       | 3/29/2026 | Lab                          | Pass   | AES-based encryption/decryption test completed successfully.                | 3/29/2026              |
| 20       | 3/29/2026 | Lab                          | Pass   | Before/after hashes matched for transferred data.                           | 3/29/2026              |
| 30       | 3/29/2026 | Lab                          | Pass   | Test evidence and pass/fail recording were documented for the demonstration | 3/29/2026              |
| 40       | 3/29/2026 | Lab                          | Pass   | Software workflow executed successfully during the demo.                    | 3/29/2026              |

**Figure 10:** Standards Test Results **Source:** Author

| Constraint | Test Date | Test and Evaluation Location | Result               | Notes                                                                                                                                            | Date Passed (Accepted) |
|------------|-----------|------------------------------|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| 10         | 4/7/2026  | Lab                          | 7.1 Watts            | Used a smart plug to measure instead of a multimeter                                                                                             | 4/7/2026               |
| 20         | 4/2/2026  | Lab                          | 71.5 C               | Tested with Raspberry Pi CPU temp reader                                                                                                         | 4/2/2026               |
| 30         | 4/2/2026  | Lab                          | 0 db                 | No additional environment noise while running. Used phone db reader.                                                                             | 4/2/2026               |
| 40         | 3/29/2026 | Lab                          | .jpeg                | The file type of the images, before and after being sent, visually inspected in file explorer is .jpeg.                                          | 3/29/2026              |
| 50         | 3/29/2026 | Lab                          | 91.4 cm <sup>3</sup> | This value exceeds our original value for the UWB transceivers, but due to hardware constraints the volume of the Raspberry Pi will be accepted. | 3/29/2026              |
| 60         | 3/29/2026 | Lab                          | Pass                 | We only used laptops running Windows.                                                                                                            | 3/29/2026              |
| 70         | N/A       | N/A                          | Not tested           | With hardware constraints, we no longer used UWB.                                                                                                | N/A                    |
| 80         | 3/29/2026 | Lab                          | Pass                 | Based on the .pkg files and the code we can visually see the image files are encrypted.                                                          | 3/29/2026              |
| 90         | 3/29/2026 | Demo Room                    | Pass                 | All testing and demo was done indoors in lab or demo room                                                                                        | 3/29/2026              |

**Figure 9:** Constraints Test Results

**Source:** Author

## **3.2 Critical Milestones**

**2-2-26 – Red circle detection for images work**

**2-20-26 - Able to communicate and transfer data over Ethernet**

**2-16-26 - Webserver (front end and back end) has all the functions needed for the project**

**2-20-26 – Raspberry Pi and laptop have hashing, and encryption set up for both communication and data transferring**

**3-24-26 – All the different software and hardware pieces work on the mission and are correctly tied together to allow for the mission to pass.**

## **3.3 Fabrication Resources**

### **Hardware Needed:**

- Raspberry Pi 4
- Ethernet Cable
- Laptop

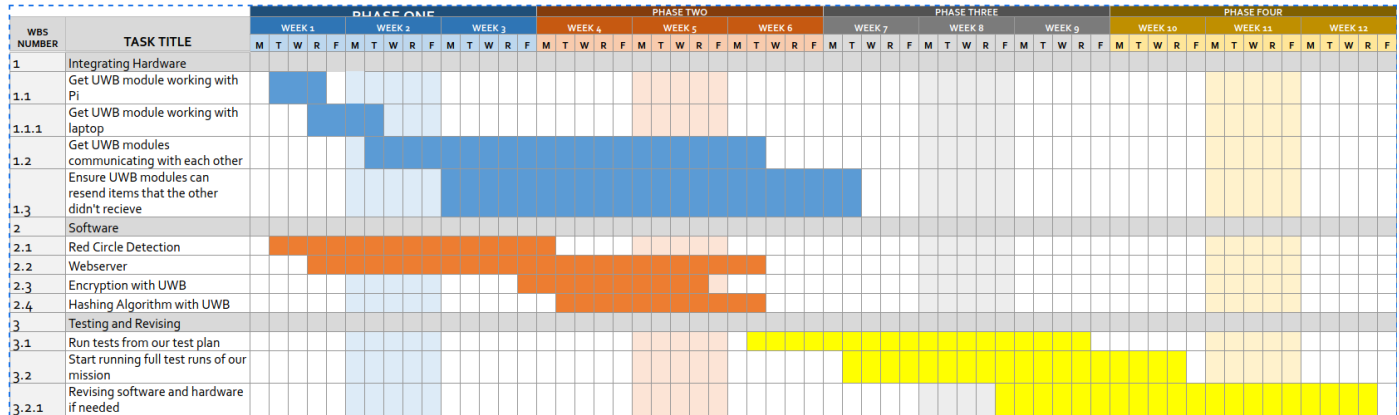
### **Software Needed:**

- Python
- OpenCV library
- Angular framework
- FastAPI framework
- AES encryption
- SHA-512 hashing
- Docker

## **3.4 Testing Resources**

- Laser heat gun
- Ruler
- Spectrum analyzer
- Terminal
- IDE (vscode)
- Multi-meter
- Decibel/sound-level meter

## 3.5 Schedule



**Figure 11: Gantt Chart**

**Source:** Author

## 3.6 Tasks

| Task ID                                                   | Start Date | End Date | Who                                | Criteria                                                                                    |
|-----------------------------------------------------------|------------|----------|------------------------------------|---------------------------------------------------------------------------------------------|
| Bill of Materials                                         | 11-1-25    | 12-6-25  | Mark Harshbarger                   | Email bill of materials is sent to Billy Kelly                                              |
| Get Raspberry Pi and laptop communicating with each other | 1-26-26    | 2-20-26  | Dylan Brumbaugh & Andrew Shoemaker | Pi and Laptop can send items over Ethernet                                                  |
| Red Circle Detection                                      | 1-12-26    | 2-2-26   | Mark Harshbarger & Blake Colegrove | Detection identifies images with red circles with 100% accuracy                             |
| Webserver - Frontend                                      | 1-16-26    | 2-16-26  | Dylan Brumbaugh                    | Frontend is fully operational with all the pages and buttons needed                         |
| Webserver-Backend                                         | 1-16-26    | 2-16-26  | Dylan Brumbaugh & Mark Harshbarger | Backend is fully operational with two-factor-auth                                           |
| Encryption                                                | 2-9-26     | 2-20-26  | Andrew Shoemaker                   | Data being transferred is fully encrypted and must be decrypted on the other end to be read |
| Hashing                                                   | 2-12-26    | 2-20-26  | Andrew Shoemaker                   | Data sent is hashed correctly to ensure full data transfer                                  |
| Run tests from our test plan                              | 2-20-26    | 3-20-26  | Team                               | Tests successfully ran based on test plan                                                   |
| Start running full test runs                              | 3-10-26    | 3-24-26  | Team                               | Full run tests of project successful                                                        |
| Revising software and hardware if needed                  | 3-19-26    | 4-21-26  | Team                               | If needed, adjust hardware and software                                                     |

**Figure 12:** Task Schedule    **Source:** Author

### 3.7 Schedule Risk

A realized risk that this project underwent is that our ordered parts may not arrive within a reasonable time. To answer this, we transitioned from UWB design to an Ethernet design which allow us to stay on track to complete the project. Another risk is if a task is taking longer than expected. To mitigate this risk, we will allow the team members working on the task to be able to change, i.e. to help with the task taking longer than expected or to take on a different task for the team member working on the overdue task. This ensures that if one task is taking longer than expected, we should hopefully have other tasks getting worked on in accordance with the start and end dates.

### 3.8 Cost

| Part               | Amount  | Cost    | Total Cost |
|--------------------|---------|---------|------------|
| Raspberry Pi 4     | 1       | \$30.00 | \$30.00    |
| Ethernet Cable     | 1       | \$10.00 | \$10.00    |
| Estimated Shipping | \$25.00 |         |            |
| Subtotal           | \$40.00 |         |            |
| Total              | \$65.00 |         |            |

**Figure 13:** Parts List

**Source:** Author

### 3.9 Cost Risk

\*Table: x axis is severity, y axis is likelihood

#### Risk 1 - Delay of schedule with ordered parts not coming in on time

Likelihood: moderate

Severity: low to moderate

Mitigation: pick parts as early as possible to ensure we get the parts as early as possible

Unmitigated:

|   | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| 5 |   |   |   |   |   |
| 4 |   |   |   |   |   |
| 3 |   | x |   |   |   |
| 2 |   |   |   |   |   |
| 1 |   |   |   |   |   |

Mitigated:

|   | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| 5 |   |   |   |   |   |
| 4 |   |   |   |   |   |
| 3 |   |   |   |   |   |
| 2 |   | x |   |   |   |
| 1 |   |   |   |   |   |

**Figure 14:** Risk 1 Assessment Matrix

**Source:** Author

### 3.10 Technical Risk

\*Table: x axis is severity, y axis is likelihood

#### Risk 2 – Shock or burns when soldering pins and assembling hardware

Likelihood: low

Severity: moderate to high

Mitigation: wear personal protective equipment (PPE) and follow safety precautions

Unmitigated:

|   | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| 5 |   |   |   |   |   |
| 4 |   |   |   |   |   |
| 3 |   |   |   |   |   |
| 2 |   |   |   | x |   |
| 1 |   |   |   |   |   |

Mitigated

|   | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| 5 |   |   |   |   |   |
| 4 |   |   |   |   |   |
| 3 |   |   |   |   |   |
| 2 |   |   |   |   |   |
| 1 |   | x |   |   |   |

**Figure 15:** Risk 2 Assessment Matrix

**Source:** Author

### Risk 3 – Fry parts due to faulty assembly, bad code, or too much power

Likelihood: low

Severity: moderate to high

Mitigation: Carefully consider code before execution (ex. Loops to not be infinite, or stack overflows, etc...), carefully monitor and manage power input to components, and ensure assembly (especially soldering) is done correctly.

Unmitigated:

|   | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| 5 |   |   |   |   |   |
| 4 |   |   |   |   |   |
| 3 |   |   |   |   |   |
| 2 |   |   |   | x |   |
| 1 |   |   |   |   |   |

Mitigated:

|   | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| 5 |   |   |   |   |   |
| 4 |   |   |   |   |   |
| 3 |   |   |   |   |   |
| 2 |   | x |   |   |   |
| 1 |   |   |   |   |   |

**Figure 16:** Risk 3 Assessment Matrix

**Source:** Author

## **4. Team Résumé**

# Blake Colegrove

513.806.7316 | blake.m.colegrove@gmail.com | Dayton, Ohio

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## Education

### Bachelor of Science in Computer Science

Wright State University, Dayton, Ohio

May 2026

GPA: 3.7

## Technical Skills

- **Programming Languages:** Java, LC3 Assembly, BASH script, SQL, PICK Basic, COBOL, C++
- **Operating Systems:** Windows 7, Windows 10, Windows 11, Linux
- **Other Useful Skills:** Database Organization, Linux command line, PowerShell command line, Microsoft Office Products

## Relevant Coursework

- Computer Science
- Discrete Structures and Algorithms
- Operating System Concepts and Usage
- Computer Organization
- Introduction to Software Development
- Comparative Languages
- Introduction to Databases and Modeling
- Technical Communications for Engineers and Computer Scientists
- Calculus I and II
- Theoretical Foundations of Computing

## Projects

### Sum of Squares (Assembly)

November 2023

- Utilized a runtime stack
- Called subroutines

### GUI Game (Java/Java Swing)

March 2023

- Utilized Java Swing to create an interactive GUI
- Complete Java Project

### Password Cracker (Java)

April 2023

- Utilized multi-threading for efficiency
- Unzipped a zip file by cracking the password

## Work Experience

### Software Development Intern | Reynolds and Reynolds

May 2024 – August 2024

- Modified existing code to add new features
- Debugged and fixed errors occurring within a program's code files

### Apparel Outfitter, Cashier | Bass Pro Shops / Cabela's

January 2022 – Present

- Use a database system to look up inventory or item information.
- Interact with customers and use a cash register system to cash them out.
- Organize floor displays based on knowledge of product, sales records, and product purpose.
- Create sale signs using design software.
- Take phone calls to answer customer questions.

## Leadership Experience

- Air Force ROTC Field Training June 2024 – July 2024
- Flying Aces Club Treasurer August 2023 – Present
- AS200 Class Leader January 2024 – May 2024

# Dylan E. Brumbaugh

(937) 503-3893 | [brumbaugh.20@wright.edu](mailto:brumbaugh.20@wright.edu) | Fairborn, OH

<https://www.linkedin.com/in/dylanbrumbaugh/>

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## EDUCATION:

### Bachelor of Science in Computer Engineering

Wright State University, Dayton, OH

Aug. 2022-present

GPA: 3.4

### Relevant Coursework:

- Data Structures and Algorithms
- Digital System Design
- Software Lifecycle
- OS Design
- Operating Systems
- Analog Circuits
- Microprocessor Embedded Systems

## TECHNICAL SKILLS:

**Languages:** HTML, CSS, JavaScript, SQL, C++, C, Java, Python

**Software Tools:** Visual Studio, Docker, DBeaver, Git, Linux Terminal, MATLAB, Arduino IDE, Clion, Logisim, Vivado

**Computer:** Putty, Microsoft Excel, Word, Presentation, Teams, OneNote

## RELEVANT EXPERIENCE:

### Software Engineer Intern, MZA Associates Corporation

May. 2025 – Present

- Contributed to full-stack development using Angular and Flask, enabling smooth communication between frontend and backend.
- Implemented responsive UI components and reusable layouts to improve user experience and maintainability
- Fixed bugs across the frontend and backend, improving application stability and performance
- Collaborated with teammates to review code, troubleshoot issues, and deliver updates to customers.

## RELEVANT PROJECTS:

### Software Development

Fall. Semester 2024

- Collaborated with classmates to complete a frontend to backend web application using MariaDB, spring boot, and angular.

### Digital System Design – Small Scale Microprocessor

Fall. Semester 2024

- Designed and built a small-scale microprocessor in Logisim, that could do basic functions such as addition, subtraction and store the data in memory.

### OS Design

Fall. Semester 2025

- Performed tasks throughout semester on small-scale OS system that implemented simple FAT file systems.

## ADDITIONAL EXPERIENCE:

Tree Service Groundman

June 2021 – Aug 2022

Concrete Laborer

June 2023 – Aug 2024

# Mark Harshbarger

Sidney, OH | 937-638-5051 | [markharshbarger25@gmail.com](mailto:markharshbarger25@gmail.com) | <https://github.com/markharshbarger>

## EDUCATION:

|                                                |          |
|------------------------------------------------|----------|
| <b>Bachelor of Science in Computer Science</b> | May 2026 |
| Wright State University, Dayton, OH            | GPA: 4.0 |

## SKILLS:

*Programming Languages:* Java, Python, JavaScript  
*Scripts / Standards:* Bash Scripting, SSH  
*Operating Systems:* Linux, Windows  
*Software Tools:* Visual Studio Code, GitHub, Docker, GitLab  
*Build Frameworks:* Gradle  
*Databases:* PostgreSQL

## RELEVANT COURSEWORK:

- |                               |                                  |                                      |
|-------------------------------|----------------------------------|--------------------------------------|
| • Operating Systems and Usage | • Data Structures and Algorithms | • Discrete Structures and Algorithms |
| • Intro to Cyber Security     | • Computer Organization          | • Linear Algebra                     |
| • Machine Learning            | • Computer Science (Java)        |                                      |

## PROFESSIONAL EXPERIENCE:

|                                                                                                                                                                                                                                                                                                                                      |                            |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| <b>Software Engineering Intern</b> , nVoq Incorporated, Boulder, CO                                                                                                                                                                                                                                                                  | May 2025-August 2025       |
| <ul style="list-style-type: none"><li>Fine-tuned LLM’s using AWS Sagemaker</li><li>Accelerated LLM training by 3.5× by optimizing GPU utilization</li><li>Collaborated with NLP Engineers to preprocess data for training</li><li>Developed k6 tests utilized in testing processes</li><li>Used boto3 and pandas in Python</li></ul> |                            |
| <b>Engineer</b> , Buckeye Electrical Products Inc, Botkins, OH                                                                                                                                                                                                                                                                       | February 2020 – April 2023 |
| <ul style="list-style-type: none"><li>Optimized construction of wire harnesses by utilizing SolidWorks to design custom 3D models</li><li>Operated 3D printers to manufacture 3D models</li><li>Developed electrical tests for wire harnesses using IVISion Software</li></ul>                                                       |                            |

## WORK EXPERIENCE:

|                                                                                                                      |                                            |
|----------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| <b>Construction Worker</b> , JBecker Construction, Piqua, OH                                                         | May 2023-August 2023, May 2024-August 2024 |
| <ul style="list-style-type: none"><li>Renovated homes and apartments by painting, drywalling, and flooring</li></ul> |                                            |

## PROJECTS:

|                                                                                                                                                                                          |                  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| <b>Blackjack</b>                                                                                                                                                                         | Spring 2024      |
| Course Project   <a href="https://github.com/markharshbarger/1181/tree/main/projects/project2blackjack">https://github.com/markharshbarger/1181/tree/main/projects/project2blackjack</a> |                  |
| <ul style="list-style-type: none"><li>Used Swing in Java to make a Blackjack Game</li><li>Implemented observers to help separate game logic and GUI</li></ul>                            |                  |
| <b>Movie List Database</b>                                                                                                                                                               | Winter 2024-2025 |
| Individual Project   <a href="https://github.com/markharshbarger/Movie-list-fetcher-psql">https://github.com/markharshbarger/Movie-list-fetcher-psql</a>                                 |                  |
| <ul style="list-style-type: none"><li>Set up a PostgreSQL database</li><li>Used Psycopg2 in Python to parse through movies on home server and add to database</li></ul>                  |                  |

## Andrew Shoemaker

937-489-6662 | [shoemaker.59@wright.edu](mailto:shoemaker.59@wright.edu) | [ashoemaker6557.github.io](https://github.com/ashoemaker6557) | [LinkedIn](#) | Anna, OH

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### PROFESSIONAL SUMMARY:

A dedicated web development professional currently pursuing a bachelor's degree in cybersecurity, combining technical expertise in web development with a strong focus on security to deliver innovative and secure solutions.

### EDUCATION:

#### Bachelor of Science in Information Systems & Cyber Security

Wright State University, GPA: 4.0

May 2026

Dayton, OH

#### Associate of Business in Interactive Media

Edison State Community College, GPA: 3.913

May 2023

Piqua, OH

### COURSEWORK:

- Security Attacks and Defenses
- Secure Computing Practices
- Introduction to Cyber Security
- Introduction to Digital Forensics
- Web Development
- Mobile Application Development
- Introduction to PC Networking
- Database Management
- Java Programming
- Python Programming

### PRIMARY SKILLS:

- Problem-Solver
- Team Player, Sociable
- Good Time-Management Skills
- Fast Learner

### TECHNICAL SKILLS:

- HTML & CSS
- JavaScript, Java, PHP, Kotlin & Python
- Android Studios & Visual Studio Code
- Ubuntu

### EXPERIENCE:

#### Crown Equipment Corporation, New Bremen, Ohio

Jan 2025 – Aug 2025

##### Cyber Security Co-op

- Tested and validated flagged vulnerabilities to distinguish true risks from false positives.
- Developed Python script to help automate security reporting tasks.

#### RWS, Remote

May 2024 – Dec 2024

##### Search Quality Rater

- Assess the relevance and quality of search engine results to ensure they meet the user's query intent.
- Follow detailed guidelines to score the quality of web pages based upon factors such as authority, trustworthiness, and content relevance.

#### Walmart, Sidney, Ohio

June 2022 – Aug 2023

##### O/N Stocking

- Stocked Merchandise on store shelves.
- Moved several 100 lbs. of produce and meat to freezer nightly.

### Projects:

#### Scribe's Desk Website,

- Used HTML, CSS, JavaScript, and PHP
- Website interacts with a MySQL database allowing users to save their writings

#### Blackjack Java GUI

- Worked in Agile sprints to develop a fully functioning Java GUI to play the game Blackjack.

## Thaer Allan

West Chester, OH | 513-848-9693 | thaerallan42@gmail.com

### EDUCATION:

#### **Bachelor of Science in Electrical Engineering**

Wright State University, Dayton, OH

May 2026 | GPA: 3.2

### SKILLS:

**Engineering Tools:** MATLAB, LTSpice, Xilinx Vivado, Studio 5000, Multisim

**Programming:** C++

**Hardware:** PLC Systems, Sensors, Motors, Oscilloscope, Function Generator

**Maintenance:** Conveyor Systems, Troubleshooting, Preventive Maintenance

**Soft Skills:** Leadership, Communication, Technical Documentation

### RELEVANT COURSEWORK:

- Control Systems • Circuits I & II • Digital Logic • Microprocessors
- Signals & Systems • Electromagnetics • PLC Programming

### PROFESSIONAL EXPERIENCE:

#### **Maintenance Technician, Amazon – Monroe, OH**

October 2025 – Present

- Preventive maintenance on conveyors, motors, and electrical systems.
- Troubleshoot electrical/mechanical issues to reduce downtime.
- Replace faulty sensors, belts, and drives following safety protocols.

#### **Mechatronic & Robotic Technician, Amazon – Vandalia, OH**

July 2025 – October 2025

- Maintained automated handling systems and completed diagnostics.
- Performed system calibration and preventive maintenance.

#### **Process Assistant, Amazon – Springdale, OH**

August 2024 – July 2025

- Led workflow coordination and daily operations.
- Trained new associates on safety and process improvements.
- Assisted managers with performance tracking and documentation.

### PROJECTS:

#### **Garage Door Control PLC Program (EE 4120L)**

- Built automated door system using Allen-Bradley PLCs and Studio 5000.

**Common-Emitter NPN Amplifier (EE 3310L)**

- Built and analyzed a 2N3904 amplifier; measured gain and frequency response.

**7×7 Booth Multiplier (Digital IC Design)**

- Designed VHDL multiplier, validated with simulation, optimized for FPGA.